

Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** across multiple categories to guide strategic business decisions.



Dataset at a Glance

3,900

Total Purchases

Rows in the dataset

18

Columns

Features tracked per transaction

37

Missing Values

Only in Review Rating column

50

Locations

Unique customer locations

Key features include customer demographics, purchase details, and shopping behavior (discounts, promo codes, shipping type, review ratings). Missing Review Rating values were imputed using the median rating per product category.

Data Preparation in Python

→ Load & Explore

Imported via `pandas`; used `df.info()` and `.describe()` for structure and summary stats.

→ Clean & Standardize

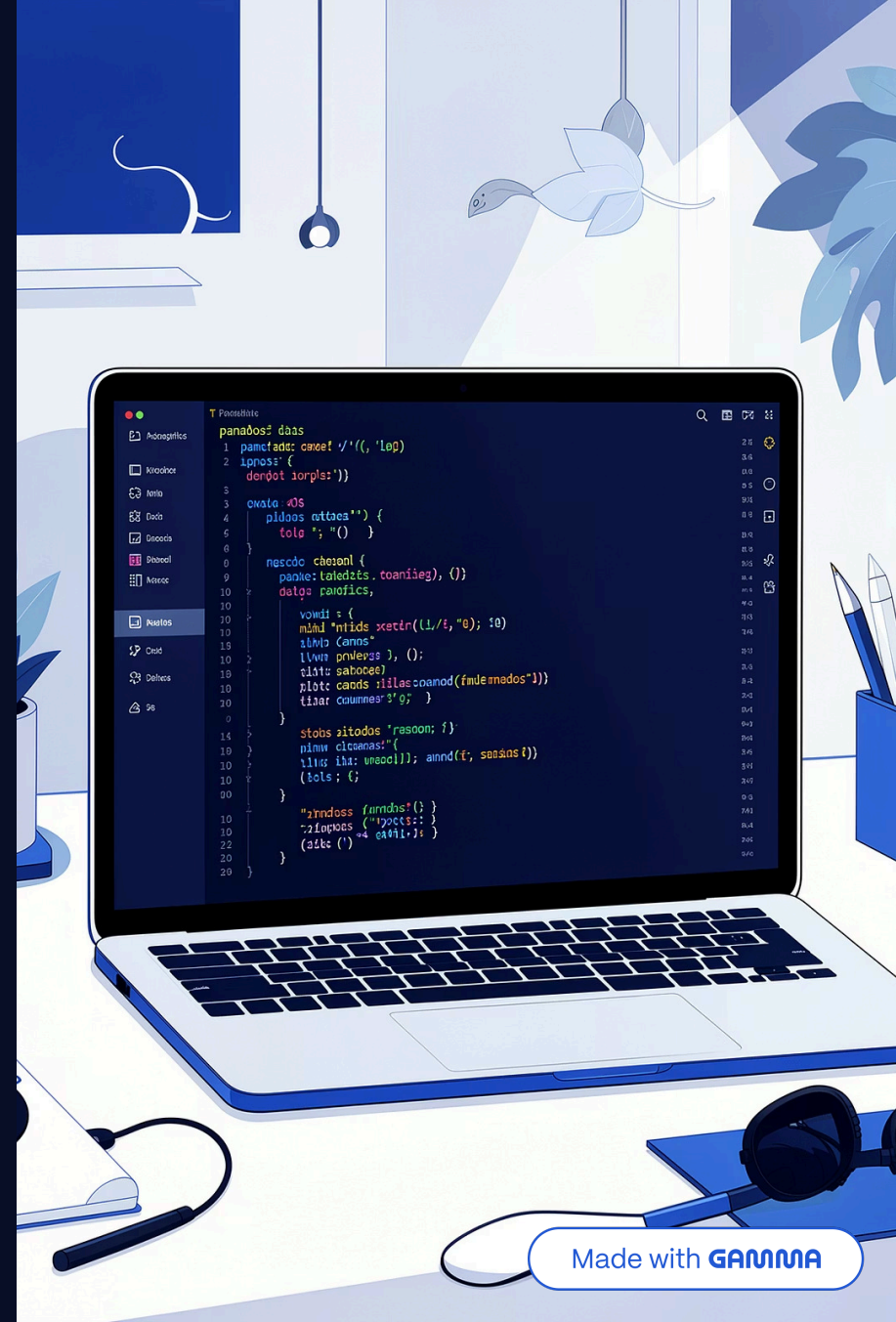
Imputed missing Review Ratings by category median; renamed columns to `snake_case`.

→ Feature Engineering

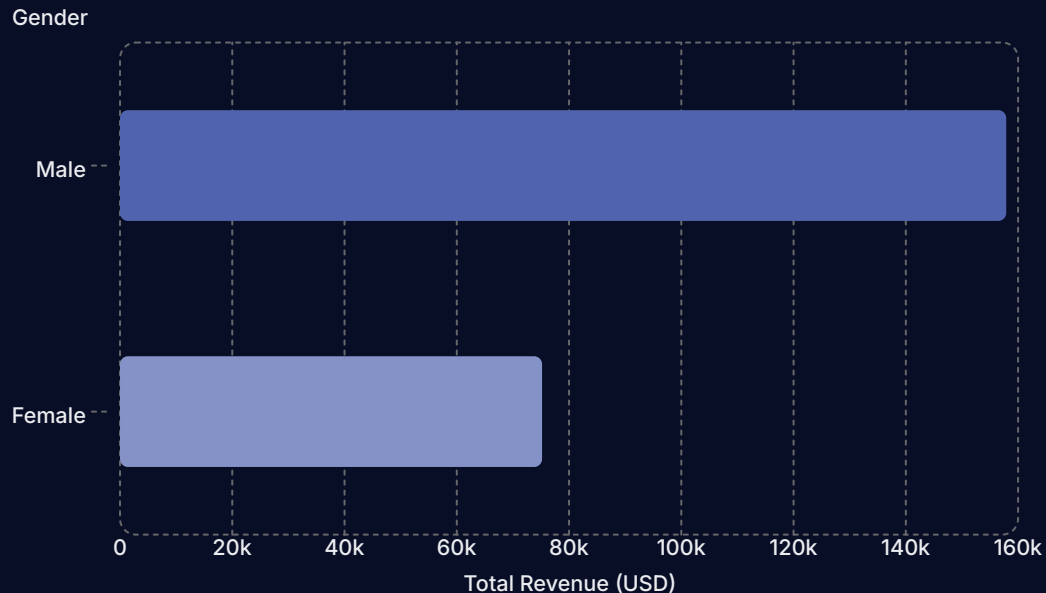
Created `age_group` (binned ages) and `purchase_frequency_days`; dropped redundant `promo_code_used`.

→ Database Integration

Loaded cleaned DataFrame into PostgreSQL for SQL analysis.



Revenue by Gender & Customer Segmentation



Gender Revenue Split

Male customers drive **67.74%** of total revenue (2,652 customers) vs. Female at **32.26%** (1,248 customers).

Customer Segments

Based on previous purchases:

- **Loyal:** 3,116 customers
- **Returning:** 701 customers
- **New:** 83 customers

Product Performance & Discount Insights

Top 5 Products by Avg Review Rating

Item	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Handbag	3.78

Highest Discount Usage Products

Item	Discount %
Hat	50.00%
Sweater	48.17%
Coat	49.07%
Pants	47.37%
Dress	45.18%

Shipping, Subscribers & Repeat Buyers



Express vs. Standard Shipping

Express shipping correlates with slightly higher average purchase amounts across all categories – Outerwear shows the largest gap (\$56 vs. \$52).

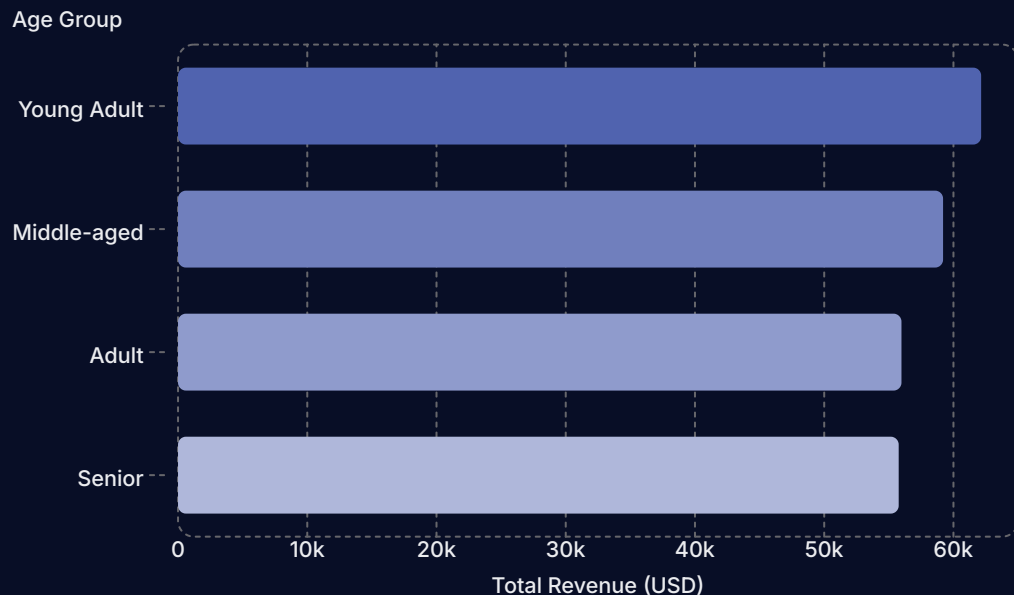
Subscriber Spending

Subscribers (1,053) and non-subscribers (2,847) both average **\$59** per purchase – subscription status doesn't affect spend per order.

Repeat Buyers

Among customers with >5 previous purchases: **958 are subscribers** vs. **2,518 non-subscribers**.

Revenue by Age Group & Top Products per Category



Top 3 Products per Category

- **Accessories:** Jewelry (171), Belt (161), Sunglasses (161)
- **Clothing:** Blouse (171), Pants (171), Shirt (169)
- **Footwear:** Sandals (160), Shoes (150), Sneakers (145)
- **Outerwear:** Jacket (163), Coat (161)

Young Adults generate the highest revenue at **\$62,143**, making them the most valuable demographic segment.

Power BI Dashboard Overview

An interactive dashboard with filters for subscription status, gender, category, and shipping type.

3.9K

Avg Customers

\$59.76

Avg Purchase Amount

3.75

Avg Review Rating

Visualizations include a donut chart (27% subscribers vs. 73% non-subscribers), bar charts for revenue by category (Clothing \$104K, Accessories \$74K, Footwear \$39K, Outerwear \$19K), and revenue by age group.

Key Dashboard Insights

73% Non-Subscribers

Large opportunity to grow the subscription base with targeted incentives.

Young Adults Lead Revenue

Highest-revenue age group – prioritize age-specific campaigns and product recommendations.

Clothing Dominates

\$104K in revenue – ensure stock availability and feature Clothing prominently in promotions.

Low Senior Revenue

Investigate with targeted offers or senior-specific product lines to close the gap.



Business Recommendations

01

Increase Subscription Rate

Offer exclusive discounts or early access to convert the 73% non-subscribers.

02

Target Young Adults

Create age-specific campaigns – the top revenue-generating segment.

03

Strengthen Clothing Category

Prioritize inventory and promotions for the \$104K top performer.

04

Build Customer Loyalty

Reward repeat buyers to move them into the "Loyal" segment and boost retention.